

soc 111 - su 2024 - 02a

GR

2024-08-13

Load libraries

```
library(igraph)
```

```
##
## Attaching package: 'igraph'

## The following objects are masked from 'package:stats':
##
##   decompose, spectrum

## The following object is masked from 'package:base':
##
##   union
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x lubridate::%--%()      masks igraph::%--%()
## x dplyr::as_data_frame() masks tibble::as_data_frame(), igraph::as_data_frame()
## x purrr::compose()       masks igraph::compose()
## x tidyr::crossing()       masks igraph::crossing()
## x dplyr::filter()         masks stats::filter()
## x dplyr::lag()            masks stats::lag()
## x purrr::simplify()       masks igraph::simplify()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Components

- Can you walk from UCLA to downtown LA?
- Given enough time, could you walk from UCLA to Harvard?

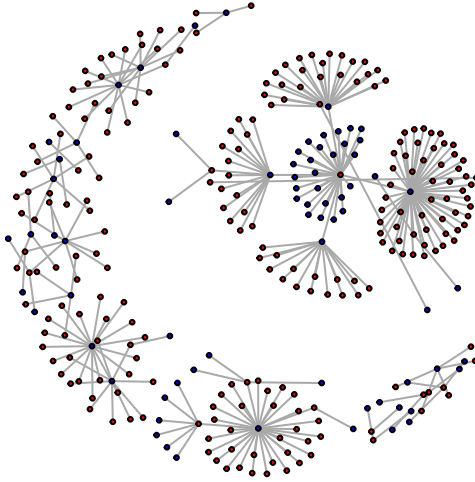
- What about UCLA to Catalina?

85.5% of the land area of the USA is in the lower 48.



Japan has 14,000 islands, but Honshu alone is 60.3% of the land area of Japan.

```
iliad.g <- read_graph("iliad_killed.net", c("pajek")) %>%  
  as_undirected()  
plot.igraph(iliad.g,  
  vertex.label=NA,  
  layout = layout_with_kk,  
  vertex.size = 2)
```



```
iliad.comp <- components(iliad.g)
```

```
iliad.comp$csizes # sizes of components
```

```
## [1] 45 10 143 12 14 5 3 19 7 4 6 2 4 2 5 3 11 3 2
## [20] 3 2 2 4
```

```
iliad.comp$csizes %>% length() # number of components
```

```
## [1] 23
```

```
sort(iliad.comp$csizes, decreasing = T)[1] # number of nodes in giant component
```

```
## [1] 143
```

```
sort(iliad.comp$csizes, decreasing = T)[1] / vcount(iliad.g) # giant component as % of whole
```

```
## [1] 0.4598071
```

```
iliad.comp$membership # component membership by node
```

##	Abas	Ablers	Acamas	AcamasAntenor
##	1	2	3	4
##	Achilles	Adamas	Adrestus_1	Adrestus_2
##	3	4	5	1
##	Adrestus_3	Aeneas	Aenius	Aesepus
##	3	1	3	6
##	Aesymnus	Agamemnon	Agastrophus	Agelaos
##	3	5	1	1
##	Agelaus	Agenor	Alastor	Alcandrus
##	3	7	8	8
##	Alcathous	Alcmaon	Amopaon	Amphiclus
##	9	3	3	10
##	Amphimachus	AmphiusMerops	AmphiusSelagus	Amphoterus
##	3	1	3	3
##	Anchialus	Antilochus	Antiphates	Antiphus
##	3	2	11	5
##	Aphareus	ApisaonPaeonia	ApisaonPhausius	Arcesilaus
##	1	12	13	3
##	Archelochus	Archeptolemos	Areilycus	Areithous
##	3	3	3	3
##	Ares	Aretaon	Aretus	Asaeus
##	1	3	14	3
##	Ascalaphus	Asius	Asius_charioteer	Asteropaeus
##	4	9	2	3
##	Astyalus	Astynous	Astypylus	Atymnius
##	15	1	3	2
##	Automedon	Autonous_A	Autonous_T	Axylus
##	14	3	3	1
##	Bathycles	Bienor	Calesius	Caletor
##	3	5	1	3
##	Cebriones	Charops	Chersidamas	Chromius1
##	3	8	8	1
##	Chromius2	Chromius3	Cleitus	Cleobulus
##	8	3	3	16
##	Clonius	CoeranusIphitus	CoeranusOfLyctus	Coön
##	7	8	3	5
##	Crethon	Croesmus	Daitor	Damasus
##	1	10	3	15
##	Dardanus	Deicoon	Democoön	Demoleon
##	3	5	8	3
##	Demouchos	Deucalion	Deïochus	Deïopites
##	3	3	1	8
##	Deïphobus	Deïpyrus	Diomedes	Diores
##	4	17	1	18
##	Dolon	DolopsClytius	DolopsLampus	Doryclus
##	1	3	17	3
##	Dresus	Dryops	Echeclus	EcheclusAgenor
##	6	3	3	3
##	Echemmon	Echepolus	EchiusLycian	Echius_A
##	1	2	3	19
##	Eioneus	Elasus	Elatus	Elephenor
##	3	3	5	7
##	Eniopeus	Ennomus	Epaltes	Epeigeus
##	1	8	3	3

##	Epicles	Epistor	Erylaus	Erymas1
##	3	3	3	9
##	Erymas2	Euchenor	Euippus	Euphorbus
##	3	1	3	17
##	Euryalus	Eurypylus	Glaucus	Gorgythion
##	6	13	3	3
##	Halius	Harpalion	Hector	HelenusPriam
##	8	4	3	17
##	Helenus_A	Hippodamas1	Hippodamas2	Hippolochus
##	3	8	3	5
##	Hippomachus	Hipponous	Hippothous	Hippotion
##	11	3	3	4
##	Hypeirochus	Hypeiron	Hyperenor	HypsenorDolopion
##	8	1	17	13
##	HypsenorHippasus	Hyrtius	Iamenus	Iasus
##	4	3	11	1
##	Idomeneus	Ilioneus	Imbrius	Ipheus
##	9	20	3	3
##	Iphidamas	Iphinous	Iphition	Isus
##	5	3	3	5
##	Laodamas	LaogonusBias	LaogonusOnetor	Leitus
##	3	3	4	21
##	Leocritus	Leonteus	Leucus	Lycaon
##	1	11	5	3
##	Lyco	Lycomedes	Lycophontes	Lycophron
##	20	12	3	3
##	Lysander	Maris	Mecisteus	Medon
##	3	22	23	1
##	Meges	Melanippus1	Melanippus2	MelanippusHicetaon
##	10	3	3	2
##	Melanthius	Menelaus	Menesthes	Menesthius
##	13	17	3	1
##	Menon	Meriones	Mermerus	Mnesus
##	11	4	2	3
##	Molion	Morys	Mulius1	Mulius2
##	8	4	3	3
##	Mydon1	Mydon2	Noemon	Odus
##	2	3	8	5
##	Odysseus	Oenomaus_A	Oenomaus_T	Oileus
##	8	3	9	5
##	Ophelestes1	Ophelestes2	Opheltius_A	Opheltius_T
##	3	3	3	6
##	Opites	Oresbius	Orestes_A	Orestes_T
##	3	3	3	11
##	Ormenus1	Ormenus2	Orsilochos	Orsilochus
##	3	15	3	1
##	Orus	Othryoneus	Otus	Oilean Ajax
##	3	9	23	16
##	Pandarus	Pandocus	Paris	Patroclus
##	1	3	1	3
##	Pedaeus	Pedatus	Peirous	Peisander
##	10	6	18	17
##	PeisanderAntimachus	Peneleus	Perimus	Periphas
##	5	20	3	1

##	PeriphetesCopeus	Periphetes_T	Phaestus	Phalces
##	3	3	9	2
##	Phegeus	Phereclus	Phorcys	Phylacus
##	1	4	3	21
##	Pidytes	Podes	Polites	Polydamas
##	8	17	19	23
##	Polydorus	Polyidus	Polymelus	Polypoetes
##	3	1	3	15
##	Promachus	Pronous	Prothoënor	Prothoön
##	4	3	23	3
##	Prytanis	Pylaemenes	Pylartes1	Pylartes2
##	8	17	3	3
##	Pylon	Pyraechmes	Pyrasus	Pyris
##	15	3	3	3
##	Rhesus	Rhigmus	Sarpedon	Satnius
##	1	3	3	16
##	Scamandrius	Schedius	Simoeisius	SleepingTrojan01
##	17	3	3	1
##	SleepingTrojan02	SleepingTrojan03	SleepingTrojan04	SleepingTrojan05
##	1	1	1	1
##	SleepingTrojan06	SleepingTrojan07	SleepingTrojan08	SleepingTrojan09
##	1	1	1	1
##	SleepingTrojan10	SleepingTrojan11	Socus	Sthenelaus
##	1	1	8	3
##	Stichius	Telamonian Ajax	Teucer	Teuthras
##	3	3	3	3
##	Thersilochus	Thestor	ThoasAndraemon	Thoas_T
##	3	3	18	17
##	Thoon	Thoön_A	Thoön_T	Thrasius
##	1	2	8	3
##	ThrasymedesNestor	Thrasymedes_T	Thymbraeus	TlepolemusDamastor
##	22	3	1	3
##	TlepolemusHeracles	Trechus	Trojan01	Trojan02
##	3	3	3	3
##	Trojan03	Trojan04	Trojan05	Trojan06
##	3	3	3	3
##	Trojan07	Trojan08	Trojan09	Trojan10
##	3	3	3	3
##	Trojan11	Trojan12	Trojan13	Trojan14
##	3	3	3	3
##	Trojan15	Trojan16	Trojan17	Trojan18
##	3	3	3	3
##	Trojan19	Trojan20	Trojan21	Trojan22
##	3	3	3	3
##	Trojan23	Trojan24	Trojan25	Trojan26
##	3	3	3	3
##	Trojan27	Tros	Xanthus	
##	3	3	1	

Paths

Milgram “six degrees”

```

# 5 = Achilles
# 14 = Agamemnon
# 123 = Hector
# 208 = Patroclus

shortest_paths(iliad.g, 5, 208) # Achilles to Patroclus

```

```

## $vpath
## $vpath[[1]]
## + 3/311 vertices, named, from c74ae4f:
## [1] Achilles Hector Patroclus
##
##
## $sepath
## NULL
##
## $predecessors
## NULL
##
## $inbound_edges
## NULL

```

```

shortest_paths(iliad.g, 5, 14) # Achilles to Agamemnon

```

```

## Warning in shortest_paths(iliad.g, 5, 14): At
## vendor/cigraph/src/paths/unweighted.c:444 : Couldn't reach some vertices.

```

```

## $vpath
## $vpath[[1]]
## + 0/311 vertices, named, from c74ae4f:
##
##
## $sepath
## NULL
##
## $predecessors
## NULL
##
## $inbound_edges
## NULL

```

```

iliad.comp$membership[5] # component for Achilles

```

```

## Achilles
##      3

```

```

iliad.comp$membership[208] # component for Patroclus

```

```

## Patroclus
##      3

```

```
iliad.comp$membership[14] # component Agamemnon
```

```
## Agamemnon  
##          5
```

```
diameter(iliad.g)
```

```
## [1] 6
```

```
farthest_vertices(iliad.g)
```

```
## $vertices  
## + 2/311 vertices, named, from c74ae4f:  
## [1] Alcmaon Bathycles  
##  
## $distance  
## [1] 6
```

```
shortest_paths(iliad.g, 22, 57)
```

```
## $vpath  
## $vpath[[1]]  
## + 7/311 vertices, named, from c74ae4f:  
## [1] Alcmaon Sarpedon Patroclus Hector Teucer Glaucus Bathycles  
##  
##  
## $epath  
## NULL  
##  
## $predecessors  
## NULL  
##  
## $inbound_edges  
## NULL
```

```
mean_distance(iliad.g)
```

```
## [1] 3.052478
```

```
mean_distance(iliad.g, unconnected = F)
```

```
## [1] Inf
```

<https://web.archive.org/web/20220709202804/https://research.facebook.com/blog/2016/2/three-and-a-half-degrees-of-separation/>

mean harmonic centrality

Harmonic mean is reciprocal of the mean of reciprocals.

$$H(x_1, \dots, x_n) = \left(\frac{n}{x_1^{-1} + \dots + x_n^{-1}} \right) = \frac{n}{\sum \frac{1}{x_i}}$$

For instance, for the set 1, 2, 3, 4 ...

$$H(1, 2, 3, 4) = \frac{4}{1/1 + 1/2 + 1/3 + 1/4} = 1.92$$

```
psych::harmonic.mean(1:4)
```

```
## [1] 1.92
```

Useful for things like average speed.

The igraph manual defines harmonic centrality as “the mean inverse distance [from ego] to all other vertices. The inverse distance to an unreachable vertex is considered to be zero.”

Based on Marchiori and Latora. 2000. “Harmony in a Small World” Physica A

For now, we are interested in the network as a whole, not each node, so take the (arithmetic) mean of the harmonic centrality scores.

```
harmonic_centrality(iliad.g) %>% mean()
```

```
## [1] 27.49561
```

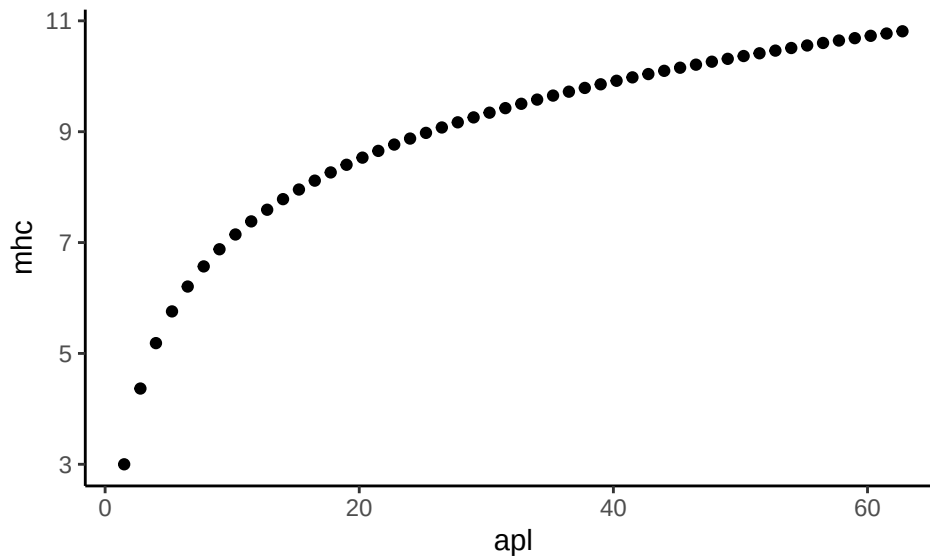
Conceptually related to average path length, but the nature of the relationship between them varies by the type of graph.

mean harmonic centrality vs average path length for rings

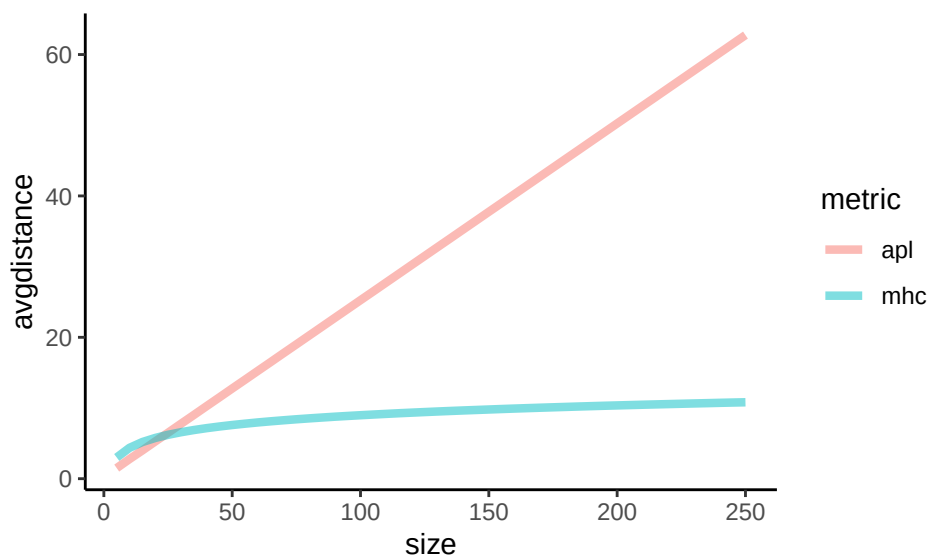
```
# you really don't need to know all this stuff with for loops and matrices
# unless you want to do a senior thesis using simulations or the like
testvalues <- seq(5,250,5)
results.mat <- matrix(nrow = length(testvalues), ncol = 3)
colnames(results.mat) <- c("size", "apl", "mhc")
j <- 1
for (i in testvalues) {
  g <- make_ring(i)
  apl <- mean_distance(g)
  mhc <- harmonic_centrality(g) %>% mean()
  results.mat[j,] <- c(i, apl, mhc)
  j <- j+1
}

results.mat %>%
  as_data_frame() %>%
  ggplot(aes(x=apl, y=mhc)) +
  geom_point() +
  theme_classic()
```

```
## Warning: 'as_data_frame()' was deprecated in tibble 2.0.0.
## i Please use 'as_tibble()' (with slightly different semantics) to convert to a
## tibble, or 'as.data.frame()' to convert to a data frame.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



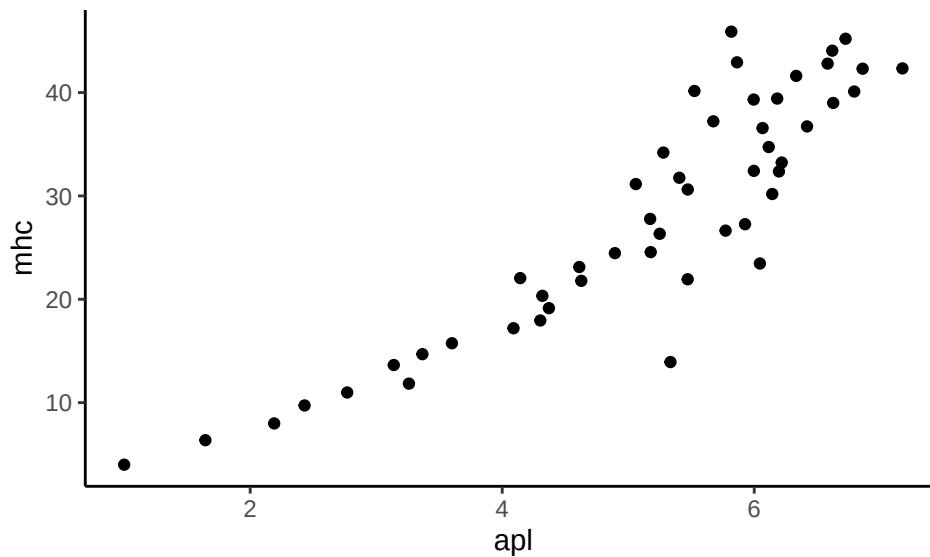
```
results.mat %>%
  as_data_frame() %>%
  pivot_longer(!size, names_to = "metric", values_to = "avgdistance") %>%
  ggplot(aes(x=size, y=avgdistance, colour = metric)) +
  geom_line(lwd=1.5, alpha=0.5) +
  theme_classic()
```



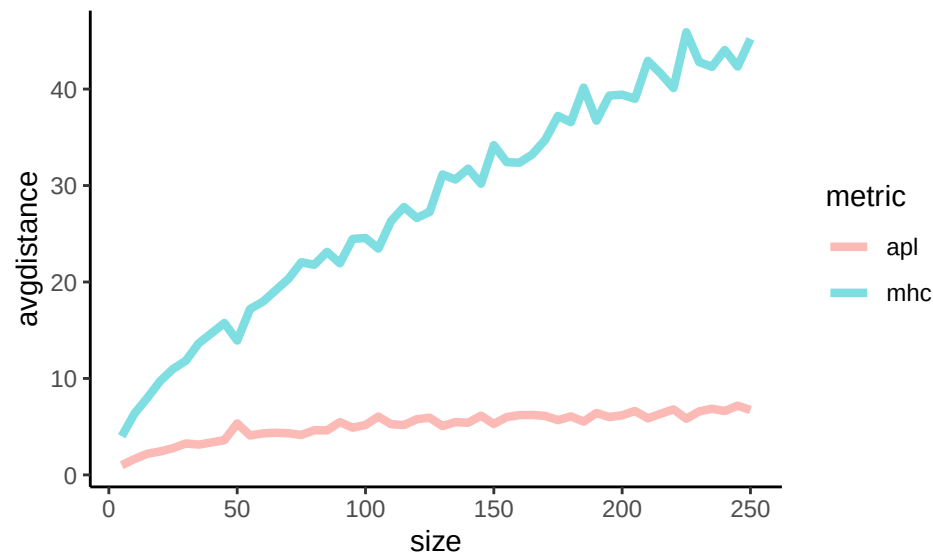
mean harmonic centrality vs average path length for Watts-Strogatz

```
ws.testvalues <- seq(10,250,5)
ws.results.mat <- matrix(nrow = length(testvalues), ncol = 3)
colnames(ws.results.mat) <- c("size","apl","mhc")
j <- 1
for (i in testvalues) {
  g <- sample_smallworld(1,i,2,.05)
  apl <- mean_distance(g)
  mhc <- harmonic_centrality(g) %>% mean()
  ws.results.mat[j,] <- c(i, apl, mhc)
  j <- j+1
}

ws.results.mat %>%
  as_data_frame() %>%
  ggplot(aes(x=apl,y=mhc)) +
  geom_point() +
  theme_classic()
```



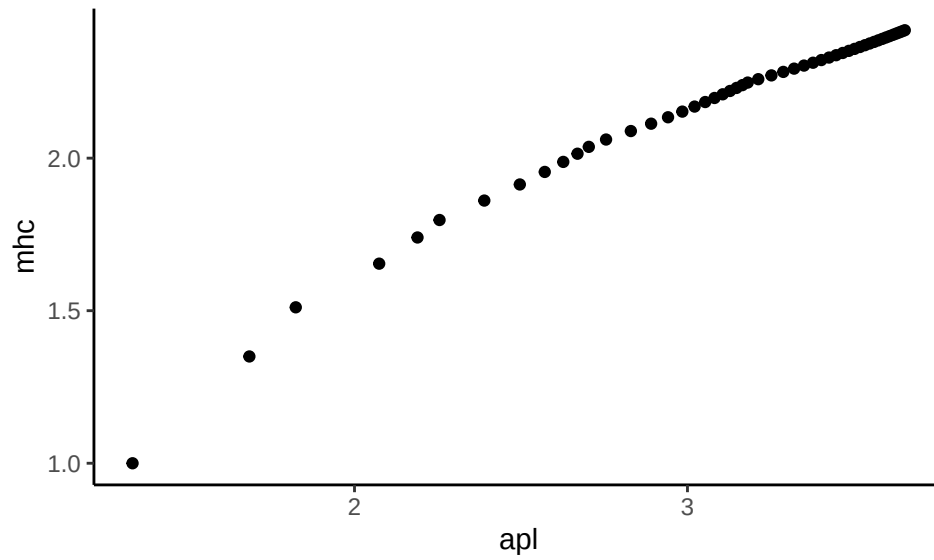
```
ws.results.mat %>%
  as_data_frame() %>%
  pivot_longer(!size, names_to = "metric", values_to = "avgdistance") %>%
  ggplot(aes(x=size, y=avgdistance, colour = metric)) +
  geom_line(lwd=1.5, alpha=0.5) +
  theme_classic()
```



mean harmonic centrality vs average path length for trees

```
tree.testvalues <- seq(10,250,5)
tree.results.mat <- matrix(nrow = length(testvalues), ncol = 3)
colnames(tree.results.mat) <- c("size","apl","mhc")
j <- 1
for (i in testvalues) {
  g <- make_tree(i,children = 2)
  apl <- mean_distance(g)
  mhc <- harmonic_centrality(g) %>% mean()
  tree.results.mat[j,] <- c(i, apl, mhc)
  j <- j+1
}

tree.results.mat %>%
  as_data_frame() %>%
  ggplot(aes(x=apl,y=mhc)) +
  geom_point() +
  theme_classic()
```



```
tree.results.mat %>%
  as_data_frame() %>%
  pivot_longer(!size, names_to = "metric", values_to = "avgdistance") %>%
  ggplot(aes(x=size, y=avgdistance, colour = metric)) +
  geom_line(lwd=1.5, alpha=0.5) +
  theme_classic()
```

